1. Types of Magnetic Materials: Dia, Para, and Ferromagnetic

Definition: Magnetic materials are substances that respond to an external magnetic field. Based on their behavior in a magnetic field, they are broadly classified into three categories by Faraday: Diamagnetic, Paramagnetic, and Ferromagnetic.

Every substance in the universe has some magnetic properties arising from the motion of electrons (orbital and spin) within its atoms. Let's study the three main types in detail:

1. Diamagnetic Materials:

- **Behavior:** They are *weakly repelled* by an external magnetic field.
- **Movement:** If placed in a non-uniform magnetic field, they tend to move from the stronger to the weaker part of the field.
- **Properties:** Their magnetic susceptibility (χ) is small and negative. Relative permeability (μ_r) is slightly less than 1. They do not retain magnetism when the external field is removed.
- **Examples:** Bismuth, Copper, Water, Gold, Silicon.

2. Paramagnetic Materials:

- **Behavior:** They are *weakly attracted* by an external magnetic field.
- **Movement:** They tend to move from the weaker to the stronger part of a non-uniform magnetic field.
- **Properties:** Their magnetic susceptibility (χ) is small and positive. Relative permeability (μ_r) is slightly greater than 1. They lose their magnetism once the external field is removed.
- **Examples:** Aluminum, Sodium, Oxygen, Platinum.

3. Ferromagnetic Materials:

- **Behavior:** They are *strongly attracted* by an external magnetic field.
- **Movement:** They rapidly move from the weaker to the stronger part of a non-uniform magnetic field.
- **Properties:** Their magnetic susceptibility (χ) is very large and positive (in thousands). Relative permeability (μ_r) is also very large ($\mu_r >> 1$). They can retain their magnetism even after the external field is removed, making them ideal for making permanent magnets.
- **Examples:** Iron, Nickel, Cobalt, and their alloys (like Alnico).

Curie Temperature: When a ferromagnetic material is heated above a certain critical temperature (called Curie temperature), it loses its strong magnetic properties and becomes paramagnetic. For Iron, this is about $770^\circ C$.

> 👉 **Hinglish Explanation:** Duniya ke saare materials magnet par alag-alag react karte hain. **Diamagnetic** materials magnet se door bhagte hain (repel hote hain, jaise paani ya copper). **Paramagnetic** thoda sa attract hote hain (jaise

aluminium). Aur **Ferromagnetic** bahut zor se chipak jate hain (jaise loha ya nickel) aur inhi se permanent magnets bante hain.

2. Magnetic Field, Lines of Force, Flux, Intensity & Magnetization

Definition: A magnetic field is the space around a magnet or a current-carrying conductor within which its magnetic influence (force of attraction or repulsion) can be experienced by other magnetic materials or moving charges.

Magnetic Lines of Force:

These are imaginary continuous closed curves used to represent the magnetic field. The tangent drawn at any point on the curve gives the direction of the magnetic field at that point.

- **Properties:**
  1. Outside the magnet, they travel from the North Pole to the South Pole.
  2. Inside the magnet, they travel from the South Pole to the North Pole (forming closed loops).
  3. Two magnetic lines of force never intersect each other.
  4. The relative closeness of the lines indicates the strength of the magnetic field (crowded = strong, far apart = weak).

Important Terms and Units:

Term	Definition	Formula / Symbol	SI Unit
Magnetic Field / Induction (B)	The force experienced by a moving charge or current element in the field. Also known as Magnetic Flux Density.	B	Tesla (T) or Wb/m^2
Magnetic Flux (Φ)	The total number of magnetic lines of force passing normally (perpendicularly) through a given surface area.	$\Phi = B \cdot A \cos\theta$	Weber (Wb)
Magnetic Intensity (H)	The degree or extent to which an external magnetic field can magnetize a material. It is independent of the medium.	$H = B / \mu_0$	Ampere/meter (A/m)
Magnetization (M or I)	The net magnetic moment developed per unit volume of a material when subjected to a magnetizing field.	$M = m / V$	Ampere/meter (A/m)

```
Magnetic Lines of Force (Bar Magnet):
          __--->__
       .-'        `-.
      /   N      S   \
     |    [======]    |  <-- Lines form closed loops
      \              /
       `-.__<--__.-'
(Outward: N to S. Inward: S to N)
```

👉 **Hinglish Explanation:** Magnetic field chumbak (magnet) ka asar dikhane wala ilaqa hai. Isko dikhane ke liye hum "Lines of force" banate hain jo North se nikal kar South me jati hain. Flux (Φ) ka matlab hai kisi area se kitni lines aar-paar ho rahi hain, iski unit Weber hoti hai. Magnetic intensity (H) wo takat hai jo kisi aur material ko magnet bana sakti hai.

3. Electromagnetic Induction (EMI) & Faraday's Laws

Definition: Electromagnetic Induction (EMI) is the phenomenon of producing an induced electromotive force (EMF) and induced current in a closed circuit or coil by changing the magnetic flux linked with it.

Discovered by Michael Faraday in 1831, EMI forms the fundamental operating principle for electrical generators, transformers, and induction motors. Without EMI, large-scale electricity generation would be impossible.

Faraday's Laws of Electromagnetic Induction:

Faraday formulated two laws based on his experiments with coils and magnets:

- **First Law (Cause of EMI):** Whenever the amount of magnetic flux linked with a closed circuit (coil) changes, an induced EMF is produced in the circuit. This induced EMF lasts only as long as the change in magnetic flux continues.
- **Second Law (Magnitude of EMI):** The magnitude of the induced EMF is directly proportional to the rate of change of magnetic flux linked with the circuit.

$$e = -N \frac{d\Phi}{dt}$$

Where:
e = Induced EMF (Volts)
N = Number of turns in the coil
$d\Phi/dt$ = Rate of change of magnetic flux (Webers/second)

Lenz's Law (The Negative Sign):

The negative sign in Faraday's formula comes from Lenz's Law. It states that the direction of the induced EMF (or induced current) is always such that it **opposes the very cause that produces it**. If you push the North pole of a magnet towards a coil, the coil's face becomes a North pole to repel the magnet. This obeys the Law of Conservation of Energy.

👉 **Hinglish Explanation:** EMI ka matlab hai bina battery ke electricity paida karna. Faraday ne bataya ki agar aap kisi coil ke paas ek magnet ko aage-peeche hilate ho, toh coil ke andar se nikalne wali lines (flux) change hoti hain, jisse current paida hota hai. Ye tab tak hi banega jab tak magnet hilta rahega. Lenz's law kehta hai ki paida hone wala current us magnet ke hilne ka virodh (oppose) karega. Generators me yahi hota hai!

4. Lorentz Force (Force on a Moving Charge)

Definition: The total force experienced by a charged particle moving through a region where both

electric and magnetic fields are present is called the Lorentz force.

When a charge q moves with a velocity v in a uniform magnetic field B, it experiences a magnetic force. (If an electric field E is also present, it experiences an electric force $F_e = qE$ as well). Let's focus on the magnetic component of the Lorentz force.

Magnetic Force: $F_m = q(v \times B)$
Magnitude: $F_m = qvB \sin\theta$

Where:
q = Magnitude of the charge (Coulombs)
v = Velocity of the charge (m/s)
B = Magnetic field strength (Tesla)
θ = Angle between the velocity vector (v) and the magnetic field vector (B)

Special Cases:

1. **When the charge is at rest ($v = 0$):** Force $F_m = 0$. A stationary charge experiences NO force in a magnetic field.
2. **When the charge moves parallel or anti-parallel to the field ($\theta = 0^\circ$ or 180°):** $\sin(0) = 0$, so $F_m = 0$.
3. **When the charge moves perpendicular to the field ($\theta = 90^\circ$):** $\sin(90) = 1$, so $F_m = qvB$ (Maximum Force). Under this condition, the charge moves in a circular path.

Direction of Force (Fleming's Left-Hand Rule):

Stretch the thumb, forefinger, and middle finger of your left hand mutually perpendicular to each other. If the **F**orefinger points to the **F**ield (B), and the **M**iddle finger points to the direction of motion of positive charge/current (v), then the **T**humb gives the direction of the **T**hrust or Force (F).

Applications:

- Particle accelerators like Cyclotrons.
- Mass spectrometers to separate isotopes based on mass.

👉 **Hinglish Explanation:** Jab koi charge (jaise electron ya proton) magnetic field ke andar daudta hai, toh uspe ek force lagta hai jise Lorentz force kehte hain. Agar charge ruka hua hai, ya magnet ki line me hi seedha ja raha hai, toh force ZERO hoga. Par agar wo 90 degree par ghoosega toh sabse zyada force lagega aur wo gol-gol ghumne lagega. Direction nikalne ke liye Fleming ka left-hand rule use hota hai.

5. Force on Current-Carrying Conductor & Rectangular Coil

Definition: A current-carrying conductor placed in an external magnetic field experiences a mechanical force because the current is essentially a collection of moving charges, each experiencing a Lorentz force.

1. Force on a Current-Carrying Conductor:

Let a straight conductor of length L carrying current I be placed in a uniform magnetic field B at an angle θ.

$$F = I(L \times B)$$

$$\text{Magnitude: } F = BIL \sin\theta$$

- Maximum force is experienced when the conductor is placed perpendicular to the field ($\theta = 90^\circ, F = BIL$).
- Zero force is experienced when placed parallel to the field ($\theta = 0^\circ$).
- The direction of this force is also given by Fleming's Left-Hand Rule.

2. Force/Torque on a Rectangular Coil in a Magnetic Field:

If we take a rectangular coil (like an armature in a motor) and pass current through it while it is suspended in a uniform magnetic field, a fascinating thing happens.

- The two parallel sides of the coil perpendicular to the magnetic field experience equal, opposite, and parallel forces.
- Because these two forces do not act along the same line, they form a **Couple**, which produces a **Torque (τ)** that causes the coil to rotate.

$$\text{Torque: } \tau = NIAB \sin\theta$$

Where:
N = Number of turns in the coil
I = Current in the coil
A = Area of the coil
B = Magnetic field strength
θ = Angle between the normal to the area of the coil and the magnetic field.

Application:

This principle is the working foundation of all **Electric Motors** (converting electrical energy into mechanical rotational energy) and Moving Coil Galvanometers.

> 👉 **Hinglish Explanation:** Current le jane wale taar (wire) ko jab hum magnet ke beech rakhte hain, toh uspar force lagta hai ($F=BIL$). Agar us taar ki ek rectangular coil (loop) bana de, toh uski ek side upar jati hai aur dusri neeche aati hai, jisse ek ghumane wala force (Torque) paida hota hai. Yahi wo basic trick hai jisse hamare ghar ke pankhe, pani ki motor, aur saare electric motors ghumte hain!

6. Moving Coil Galvanometer: Principle, Construction & Working

Definition: A Moving Coil Galvanometer (MCG) is a highly sensitive electromagnetic instrument used to detect and measure very small electric currents in a circuit.

Principle:

It works on the principle that when a current-carrying coil is suspended in a uniform magnetic field, it experiences a torque which tends to rotate the coil. The torque is directly proportional to the current flowing through it.

Construction:

- **Coil:** A rectangular or circular coil of many turns of insulated copper wire wound on a light metallic frame.

- **Magnets:** Two strong, concave-shaped permanent magnetic poles (N and S). These provide a *strong radial magnetic field* so that the plane of the coil is always parallel to the magnetic field, maximizing the torque ($\theta = 90^\circ$).
- **Soft Iron Core:** A soft iron cylinder is placed inside the coil. It increases the strength of the magnetic field and makes it radial.
- **Suspension & Springs:** The coil is suspended by a phosphor-bronze wire. A hairspring is attached at the bottom to provide a restoring torque (which brings the pointer back to zero when current stops).
- **Pointer & Scale:** A light pointer attached to the coil moves over a calibrated scale.

Working and Theory:

When current I passes through the coil, a deflecting torque acts on it:
Deflecting Torque (τ_d) = $NIAB$ (Since radial field makes $\sin\theta = 1$)
As the coil rotates, the suspension wire twists, producing a restoring torque proportional to the angle of twist (α).
Restoring Torque (τ_r) = $k\alpha$ (where k is the restoring torque per unit twist).
In equilibrium, Deflecting Torque = Restoring Torque:
$NIAB = k\alpha$
$\alpha = \left(\frac{NAB}{k}\right) I$

$\alpha \propto I$ (Deflection is directly proportional to the current)

This linear relationship allows the galvanometer to have a uniform (equally spaced) scale.

```
Moving Coil Galvanometer Diagram (Top View):
       (N) Pole                      (S) Pole
    /""""""""\     +----------+     /""""""""\
   |          |    |   Core   |    |          |
   | Radial   |====|  (Iron)  |====| Radial   |
   | Field    |    |          |    | Field    |
    \......../     +----------+     \......../
                        |
                 Pointer |
                \--------V--------/
                    0  1  2  3
```

👉 **Hinglish Explanation:** Galvanometer bahut chote current ko napne ka meter hai. Isme ek coil ko do strong cylindrical magnets ke beech rakha jata hai. Jab coil me current aata hai, toh Motor wale principle se uspar Torque lagta hai aur wo ghumti hai. Coil ke ghumne se usme laga pointer ghumta hai. Ek spring isko wapas zero par laane ka kaam karta hai. Deflection (pointer ka ghumna) direct current ke proportional hota hai ($\alpha \propto I$).

7. Conversion of Galvanometer into Ammeter and Voltmeter

Definition: A galvanometer is a sensitive device. To measure large currents (Ammeter) or large voltage differences (Voltmeter), it must be modified by connecting specific external resistors to it.

1. Conversion of Galvanometer into an Ammeter:

An ammeter is a low-resistance instrument connected in **series** in a circuit to measure current.

- **How to convert:** A very small resistance called a **Shunt (S)** is connected in **parallel** with the galvanometer.
- **Reason:** The shunt bypasses the majority of the heavy circuit current, preventing the sensitive galvanometer coil from burning out, while keeping the overall resistance of the ammeter very low.

$$S = \left(\frac{I_g}{I - I_g}\right) \times G$$

Where:
S = Shunt resistance required
G = Resistance of the Galvanometer
I_g = Current for full-scale deflection in galvanometer
I = Maximum current to be measured (Range of ammeter)
Ideal Ammeter: Should have ZERO resistance.

2. Conversion of Galvanometer into a Voltmeter:

A voltmeter is a high-resistance instrument connected in **parallel** across a component to measure potential difference.

- **How to convert:** A very high resistance (R) is connected in **series** with the galvanometer.
- **Reason:** A voltmeter must have very high resistance so that it draws negligible current from the main circuit, ensuring it doesn't alter the voltage it is trying to measure.

$$R = \left(\frac{V}{I_g}\right) - G$$

Where:
R = High series resistance required
V = Maximum voltage to be measured (Range of voltmeter)
I_g and G have the same meanings as above.
Ideal Voltmeter: Should have INFINITE resistance.

```
Ammeter Conversion:                 Voltmeter Conversion:
     Main Current I                      Total Voltage V
   ----o-----------o----               -------------------
       |           |                   |                 |
      [ G ]       [ S ]               [ G ]-----[ R ]----
       |           |                   |                 |
   ----o-----------o----               -------------------
  Galvanometer + PARALLEL SHUNT     Galvanometer + SERIES RESISTANCE
```

👉 **Hinglish Explanation:** Galvanometer se bada current nahi naap sakte warna wo jal jayega. Usko **Ammeter** banane ke liye ek bahut chota sa resistance (Shunt) uske parallel me jod dete hain, taaki extra current bypass ho jaye. Usko **Voltmeter** banane ke liye ek bahut bada resistance uske series me jod dete hain taaki circuit ka original current disturb na ho. Derivation aur numerical ke liye S aur R ke formulas yaad rakhein!

🔁 Quick Revision Notes (Exam Saver)

- **Magnetic Materials:** Dia (Repel, $\mu_r < 1$), Para (Attract weak, $\mu_r > 1$), Ferro (Attract strong, $\mu_r >> 1$).
- **Faraday's Laws (EMI):** Changing magnetic flux produces induced EMF ($e = -N d\Phi/dt$). Lenz's Law explains the minus sign (opposes the cause).
- **Lorentz Force:** Force on moving charge $F = qvB \sin\theta$. Max when $v \perp B$ ($\theta = 90^\circ$).
- **Force on Conductor:** $F = BIL \sin\theta$. Torque on coil: $\tau = NIAB \sin\theta$. Direction by Fleming's Left Hand Rule.
- **Moving Coil Galvanometer:** Based on torque on a current coil in a radial magnetic field. $I \propto \alpha$ (uniform scale).
- **Ammeter Conversion:** Connect low resistance (Shunt) in PARALLEL. (Ideal R = 0).
- **Voltmeter Conversion:** Connect high resistance in SERIES. (Ideal R = Infinite).

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